

Undergraduate Medicine

Nephrology Study Pack

CharaNas Medicine — Specialty Notes

Expanded notes with original schematic figures for revision. These are educational drawings inspired by standard undergraduate teaching themes, not copied textbook plates. Always follow your faculty curriculum and latest guidelines.

Section	Focus
1	Kidney function & AKI framework
2	Nephritic vs nephrotic
3	CKD & diabetic kidney disease
4	Electrolytes emergencies
5	Checklist & books
6	Quick pearls from these notes
7	Recommended book sources

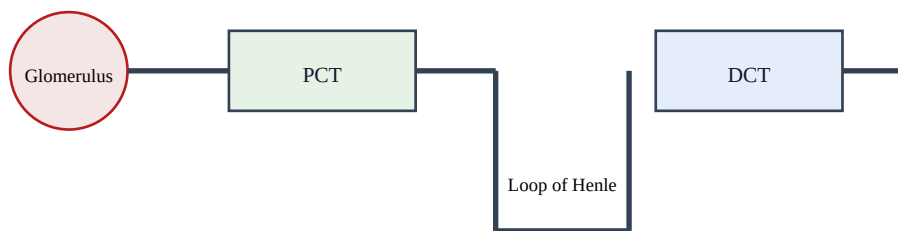
1. Kidney function & AKI framework

eGFR estimated from creatinine is the everyday index of kidney function, but acute changes, low muscle mass, and drugs affect interpretation.

AKI: rise in creatinine or drop in urine output. Classify pre-renal, intrinsic, post-renal. History of volume loss, nephrotoxins, obstruction is essential.

Muddy brown casts suggest ATN; RBC casts suggest glomerulonephritis; sterile pyuria may suggest AIN/TB depending on context.

Figure: Nephron overview (schematic)



Filtration → reabsorption/secretion → collecting system → urine

Schematic figure for learning — verify details in your recommended textbooks.

Revision prompts

- Explain this section aloud in 60 seconds as if teaching a junior student about nephrology.
- List 3 red flags that would make you escalate care urgently.
- Name 2 investigations and 2 initial management steps you would consider first.

2. Nephritic vs nephrotic

Nephritic: hematuria, hypertension, oliguria, mild–moderate proteinuria, rising creatinine.

Nephrotic: heavy proteinuria (>3.5 g/day), hypoalbuminemia, edema, hyperlipidemia; complications include thrombosis and infection.

Causes differ by age: MCD common in children; membranous/FSGS/diabetic kidney disease important in adults.

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3. CKD & diabetic kidney disease

CKD staging uses GFR categories and albuminuria. Slow progression with BP control, ACEI/ARB in albuminuric disease, SGLT2i in many diabetic/CKD pathways, diabetes control, avoid nephrotoxins.

Complications: anemia, bone mineral disorder, acidosis, hyperkalemia, volume overload, CVD risk.

Refer early when GFR declines rapidly, complications are hard to manage, or transplant/dialysis planning is needed.

Revision prompts

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- List 3 red flags that would make you escalate care urgently.
- Name 2 investigations and 2 initial management steps you would consider first.

4. Electrolytes emergencies

Hyperkalemia with ECG changes is an emergency: stabilize membrane (calcium), shift K inward, remove K, address cause.

Hyponatremia: assess volume status and tonicity; correct carefully to avoid osmotic demyelination. Hypernatremia usually means water deficit.

Acid–base paired with electrolytes unlocks many diagnoses (e.g., RTA vs diarrhea, anion-gap metabolic acidosis list).

Revision prompts

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- List 3 red flags that would make you escalate care urgently.
- Name 2 investigations and 2 initial management steps you would consider first.

5. Checklist & books

Checklist: volume status, daily weights, urine dipstick $\pm$ microscopy, creatinine/eGFR/electrolytes, bladder scan/US for obstruction, drug review (NSAIDs, ACEi, gentamicin, contrast).

Books: Brenner & Rector; Comprehensive Clinical Nephrology; Harrison's nephrology.

Revision prompts

- Explain this section aloud in 60 seconds as if teaching a junior student about nephrology.
- List 3 red flags that would make you escalate care urgently.
- Name 2 investigations and 2 initial management steps you would consider first.

6. Quick pearls

High-yield reminders packaged with this specialty in CharaNas:

1. AKI types: pre-renal, intrinsic (ATN/AIN/GN), post-renal.
2. Nephritic vs nephrotic: active sediment/HTN vs heavy protein/edema.
3. Diabetic kidney disease: ACEI/ARB cornerstone with glycemic/BP control.
4. eGFR guides drug dosing and CKD staging.
5. Hyperkalemia emergencies: ECG changes → calcium, shift, remove K+.
 - Link symptoms → anatomy/physiology → investigation → first management step.
 - Write one OSCE-style explanation for a common presentation in this specialty.
 - After reading, do the Short MCQ and Case-based Question for active recall.

7. Recommended book sources

Standard undergraduate references commonly used internationally. Use the edition your college recommends. Figures in commercial textbooks are copyrighted; use library/licensed access for full plates and photographs.

#	Reference
1	Brenner & Rector's The Kidney
2	Harrison's — Nephrology
3	Comprehensive Clinical Nephrology — Feehally

How to study with this PDF: read one section → sketch the figure from memory → answer the MCQs/cases → review mistakes → revisit the matching section.

Disclaimer: educational aid only; not a substitute for clinical guidelines, faculty teaching, or licensed textbooks.